

2. TISSUE EMBEDDING STATION

Orientation and casting of processed tissue in molten paraffin to create labelled blocks for sectioning.

SOP No.: LAB-HISTO-02	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To correctly orient processed tissue in paraffin, form a stable block and preserve diagnostic surfaces for microtomy.

PPE & Safety

- Wear lab coat and heat-resistant gloves when appropriate. Beware of hot paraffin, heated surfaces and sharp forceps. Treat tissue as potentially infectious. Keep electrical equipment and liquids separated.

Materials

- Processed tissue cassettes; embedding moulds; paraffin wax; heated forceps; cold plate; labelled cassette/base; scraper and approved cleaning materials.

Procedure

- Switch on and verify wax reservoir, work surface, forceps well and cold plate temperatures according to the validated/maker settings.
- Confirm specimen identity and cassette number before opening. Select a mould of suitable size.
- Dispense molten paraffin into the mould. Using warm forceps, place tissue at the required orientation and depth. Ensure the intended cutting surface is correctly positioned.
- Attach the labelled cassette/base and fill completely with paraffin without trapping excessive air bubbles.
- Transfer the mould to the cold plate until the wax is fully solidified.
- Remove the mould carefully. Inspect the block face and trim excess wax only as needed for secure microtome clamping.

Quality Control

- Confirm correct patient/specimen identification, orientation, complete wax infiltration and a firm, crack-free block. Re-embed if orientation is wrong or tissue is inadequately supported.

Shutdown & Cleaning

- Keep wax reservoirs at the manufacturer's standby setting or switch off as per SOP. Remove wax debris while equipment is safely cooled. Keep surfaces and forceps clean. Do not pour wax into drains.

Troubleshooting

- Tissue moves during cutting → improve orientation/support. Cracks/holes → check wax quality and embedding temperature. Block detaches from cassette → ensure correct wax filling and bonding. Uneven block → level mould/base and allow complete cooling.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.